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Understanding the Flow Behavior of Carbon Dioxide in Porous Media: Insights from Innovative Core Modeling and Two-Phase Flow Simulations

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Abstract

Understanding the flow behavior of carbon dioxide (CO_2) in porous media is essential for the success of carbon capture, utilization, and storage. Although many models have investigated CO_2 flow characteristics based on fluid properties, little attention has been paid to how the complex development and spatial distribution of pores, vugs, and fractures within porous media influence flow behavior.

In this study, we developed an innovative core modeling approach by processing the grayscale values of rock particles obtained from micro-CT scanning experiments through conversion, filtering, and interpolation. Using this method, three distinct core models were constructed: porous, vuggy, and fractured. A two-phase flow simulation technique was implemented based on the Volume of Fluid method, leveraging GPU parallel computing to improve computational efficiency. After validating the model's accuracy through experiments, we conducted simulation studies of CO_2-driven water displacement under laboratory conditions (298 K, 0.1 MPa) and supercritical carbon dioxide ($scCO_2$)-driven water displacement under reservoir conditions (354 K, 15 MPa).

The results indicate that, under laboratory conditions, CO_2 rapidly displaces water along the high-permeability zones formed by fractures in a fractured core, leading to significant water retention in smaller pores. This results in a sharp decline in the relative permeability of water and a very short duration of CO_2-water two-phase co-flow. In a vuggy core, CO_2 preferentially invades vugs with higher concentrations of mobile fluids, achieving the highest sweep efficiency and the longest two-phase co-flow duration. For a porous core, the residual water saturation and co-flow duration fall between those of the fractured and vuggy cores. Under reservoir conditions, the viscosity ratio of $scCO_2$ to water increases, and their flow follows a liquid-liquid regime. In porous and vuggy cores, the invasion efficiency of $scCO_2$ improves, prolonging the two-phase co-flow stage and displacing more water from corners. However, in fractured cores, the flow behavior of $scCO_2$ and water closely resembles that observed under laboratory conditions, predominantly flowing through fractures with limited penetration into smaller pores. This suggests that

temperature, pressure, and fluid properties have minimal impact on two-phase flow in the presence of pronounced preferential flow paths.

Unlike conventional core modeling methods, the approach we propose offers a more intuitive and detailed representation of the development and distribution of vugs and fractures within reservoir rocks. It enables the visualization of fluid mobilization on a larger scale, as well as the distribution of residual fluids at microscopic scales. This study holds significant potential for advancing microscopic mechanistic research in gas storage development, oil and gas recovery, and geological CO_2 sequestration.

Key words: Core modeling, Two-phase flow simulation, Sweep efficiency, Flow paths

Introduction

Global climate change has become an urgent environmental challenge, with direct consequences that include a rise in the global-mean temperature and an increased frequency of extreme-weather events (Boubaker et al., 2025). As the dominant greenhouse gas, atmospheric carbon-dioxide (CO_2) continues to accumulate and is widely regarded as the principal driver of global warming (Evangeline and Darwin, 2025). Carbon capture, utilization, and storage (CCUS) has therefore emerged as a pivotal technological pathway for mitigating anthropogenic CO_2 emissions (Thepsaskul et al., 2025). Nevertheless, markedly improving the storage efficiency and long-term security of CO_2 requires more than an understanding of its macroscopic migration at the reservoir scale (Zhao et al., 2025); the pore-scale mechanisms by which intricate microstructures regulate hydrodynamic processes in porous media must also be elucidated (Yang et al., 2022).

Within geological formations, CO_2 migration is controlled by multiscale multiphase-flow processes that span pore-scale displacement mechanics to reservoir-scale plume evolution (Wang et al., 2013). Previous investigations demonstrate that pore morphology and connectivity profoundly influence flow patterns and displacement efficiency. High-permeability fractures, for instance, can furnish rapid conduits that accelerate breakthrough (Zhang et al., 2023; Li et al., 2025), whereas complex pore architectures often induce microscopic trapping and non-uniform flow, thereby diminishing overall sweep efficiency (Yang et al., 2025). Although CO_2-brine displacement in homogeneous porous media has been extensively examined (Zhu et al., 2016; Bakhshian & Hosseini, 2019), the impact of heterogeneous features common to natural reservoirs-micropores, vugs, and fractures-remains underexplored. Pore-scale heterogeneity directly dictates plume migration, trapping efficiency, and storage security (Liu et al., 2015; Ma et al., 2021); bridging this knowledge gap is therefore imperative.

Advances in X-ray micro-computed tomography (micro-CT) have raised pore-scale imaging resolution to ~1 μm (Bultreys et al., 2016), providing a powerful means of probing microscopic flow processes. Nevertheless, most current studies focus on simplified pore networks or single-scale features, thus failing to capture the multiscale heterogeneity of natural rocks (Tang et al., 2020). Moreover, computational constraints restrict many pore-scale simulations to domains smaller than 1 mm^3, potentially omitting critical flow phenomena at larger scales (Yu et al., 2022).

To overcome these limitations, we combine hig-resolution micro-CT imagery with digital-rock physics to construct three representative core models—porous, vuggy, and fractured cores. Our grayscale-segmentation algorithm preserves sub-resolution pore information, faithfully reproducing complex pore architectures; when coupled with GPU-accelerated Volume-of-Fluid (VOF) simulations, it markedly expands the computational domain and enhances efficiency. Systematic CO_2–brine displacement studies are then carried out under laboratory conditions (298 K, 0.1 MPa, gaseous CO_2) and reservoir conditions (354 K, 15 MPa, supercritical CO_2) to clarify how differences in fluid properties and pore structure influence relative-permeability curves, displacement morphology, and trapping mechanisms.

The principal contributions of this work are: (1) a digital-core construction methodology capable of capturing multiscale heterogeneity; (2) a quantitative elucidation of the coupled relationship between pore characteristics and CO_2 displacement efficiency under diverse hydrodynamic conditions; and (3) a mechanistic clarification of CO_2–water two-phase transport that provides theoretical guidance and methodological support for enhancing CCUS storage security and efficiency in reservoirs with complex rock structures.

Methodology

Modeling of different core types

Grayscale values obtained from micro-CT scanning record the X-ray attenuation of rock grains, which is generally inversely proportional to porosity (Lai et al., 2018); the higher the grayscale, the denser the mineral framework and the greater the bulk density. By converting and processing the grayscale dataset, a three-dimensional dataset that captures both porosity and permeability—including nano-scale throats—can be generated. This dataset preserves the complete vuggy and fracture network and can be used to construct digital cores suitable for laboratory-scale flow analyses.

The porosity dataset (ϕ_{CT}) and permeability dataset (K_{CT}) are derived from the grayscale dataset (V_{CT}). Because the core's bulk porosity ϕ is inversely proportional to its mean grayscale value ($\overline{V_{CT}}$) of the whole core, we have:

$$\phi = \alpha_{CT} \cdot \left(1/\overline{V_{CT}}\right) \tag{1}$$

where α_{CT} is a conversion coefficient. For any voxel i scanned by micro-CT, the same inverse relation holds:

$$\left[\phi_{CTi}\right]_{i=1...N} = \alpha_{CT} \cdot \left[1/V_{CTi}\right]_{i=1...N} \tag{2}$$

Equation (2) directly yields the three-dimensional porosity dataset of the core. The experimentally measured rock permeability K is empirically correlated with porosity ϕ and the mean grayscale value $\overline{V_{CT}}$ as follows:

$$K = \beta_{CT} \cdot \left(1/\overline{V_{CTi}}\right)^2 \cdot \phi \tag{3}$$

where β_{CT} is the conversion coefficient. Likewise, the permeability K_i in any CT-scanned voxel i obeys the same transformation with respect to porosity and grayscale:

$$\left[K_{CTi}\right]_{i=1...N} = \beta_{CT} \cdot \left[\left(1/V_{CT2i}\right)^2\right]_{i=1...N} \cdot \phi_{CTi} \tag{4}$$

In the course of constructing a digital-core model from micro-CT volumes, instrument resolution and core size inevitably vary from experiment to experiment, so the voxel dimensions of the exported data cubes are seldom identical. The first—and most critical—task is therefore to prescribe a target numerical grid and resample the raw data such that the interpolated cube exactly matches this grid.

Taking the grayscale dataset as an example:

(1) Grid definition

- Fix the initial aspect ratio at 1: 2 and specify a working grid of 100 × 100 × 200 (x, y, z).
- The CT data are then interpolated onto this grid so that the resampled cube has the same voxel count as the experimental volume.

(2) Coordinate matrix

- Let N_1 be the number of CT slices.
- Construct a 100 × 100 × 200 array (Axis) and populate it with the coordinate triplets

[0, 0, 1] … [0, 0, 199], [0, 1, 0] … [99, 99, 199].

- The resulting coordinate matrix is stored as Grid.

(3) Reshaping the CT cube

- Restore the one-dimensional array Data3D to an $N_1 \times W \times H$ cube, where W and H are the slice width and height.
- Use NumPy's linspace to generate the interpolation axes:
  - C_x: 0 → 100, N_2 = 941 points;
  - C_y: 0 → 100, W = 584 points;
  - C_z: 0 → 200, H = 620 points.

(4) Interpolation

- Employ SciPy's RegularGridInterpolator to perform linear interpolation of Data3D on the (C_x, C_y, C_z) mesh.
- The resulting scipy.interpolate object retains the grid coordinates, data values and interpolation scheme.
- Sample this object at the integer (or nearest-integer) nodes given in Grid to obtain the working data cube, hereafter referred to as Data.

(5) Normalisation

Direct rendering of Data yields poor visual contrast. To emphasise rock textures, all voxel values are scaled to [0, 1]:

$$X_i = \frac{X_i - \min}{\max - \min} \tag{5}$$

The normalised array is appended as an additional column to Grid and exported.

Exactly the same workflow is applied to the permeability and porosity volumes. Visualisation with the Python package Vedo subsequently produces the digital core. Figure 1 illustrates the fractured core L27: the physical specimen (Fig. 1a), CT-grayscale slice (Fig. 1b), permeability model (Fig. 1c) and porosity model (Fig. 1d).

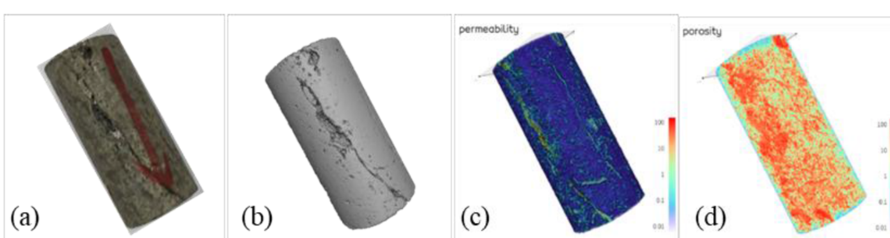

Figure 1—Digital-core model of the fractured core L27 (ϕ = 28.32 %, K = 123.51 mD)

Using the same methodology, digital cores were generated for the porous core L85 (Fig. 2) and the vuggy core L59 (Fig. 3).

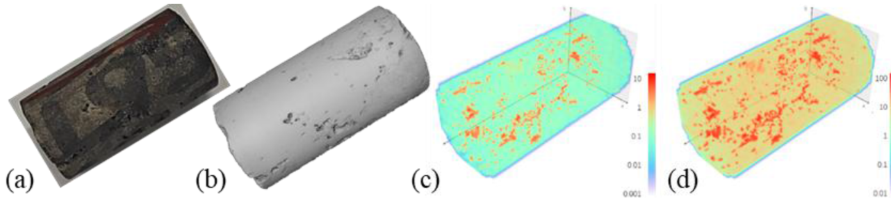

Figure 2—Digital-core model of the porous core L85 (ϕ = 7.42 %, K = 1.35 mD)

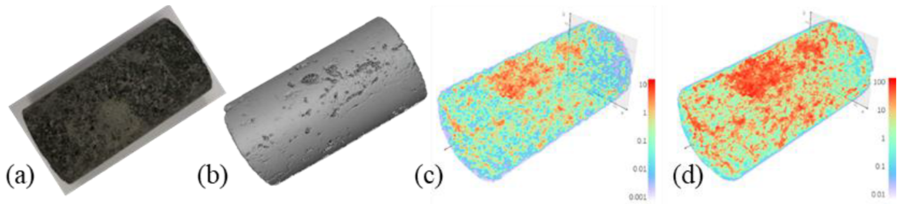

Figure 3—Digital-core model of the vuggy core L59 (ϕ = 9.33 %, K = 12.32 mD)

Carbon Dioxide-Water Two-Phase Flow Theory

The Volume-of-Fluid (VOF) method determines the velocity and pressure fields of a two-phase system by simultaneously solving the continuity and Navier–Stokes equations. A structured computational grid is first generated, on which a set of partial differential equations describing the evolution of flow parameters is formulated; numerical solutions of these equations then yield the multiphase-flow behaviour (Yang et al., 2021). Because the movement of water and CO_2 in porous rock is governed jointly by capillary, viscous and gravitational forces, we follow the CO_2–water two-phase scheme of Shams et al. (2018). The governing equations for super-critical CO_2–water flow based on the VOF framework are:

$$\nabla \cdot \boldsymbol{u} = 0 \tag{6}$$

$$\frac{\partial(\rho \boldsymbol{u})}{\partial t} + \nabla \cdot (\rho \boldsymbol{u}\boldsymbol{u}) = -\nabla p + \rho g + F_{\mathrm{c}} + F_{\mathrm{vis}} \tag{7}$$

where $\boldsymbol{u}$ is the velocity vector (m/s); ρ is the density (kg/m^3); t is time (s); p is pressure (N/m^2); g is the gravitational acceleration (m/s^2); F_{c} and F_{vis} are the interfacial and viscous forces acting on the phase boundary, respectively (N/m^3), given by:

$$F_{\mathrm{c}} = \kappa\sigma\boldsymbol{n},\ F_{\mathrm{vis}} = \nabla \cdot \left[\mu\left(\nabla \boldsymbol{u} + \nabla^T \boldsymbol{u}\right)\right] \tag{8}$$

where σ is the interfacial tension between the two phases (N/m); κ is the interface curvature; $\boldsymbol{n}$ is the unit normal vector pointing outward from the interface; and μ is the dynamic viscosity (Pa(s).

In the VOF framework, the spatial distribution of the two fluids is characterised by the volume fraction, enabling explicit interface capturing. The volume fraction α is defined as the percentage of a control volume occupied by a given phase and is expressed as (Zhu et al., 2021):

$$\frac{\partial \alpha}{\partial t} + \nabla \cdot (\alpha \boldsymbol{u}) + \nabla \cdot \left[\alpha(1-\alpha)u_{\mathrm{c}}\right] = 0 \tag{9}$$

where u_{c} is the compression velocity (m/s), taken as the maximum velocity within the interfacial region. The density and viscosity at the CO_2-water interface are therefore given by:

$$\begin{aligned} \rho_{\mathrm{m}} &= \alpha\rho_{\mathrm{w}} + (1-\alpha)\rho_{\mathrm{c}} \\ \mu_{\mathrm{m}} &= \alpha\mu_{\mathrm{w}} + (1-\alpha)\mu_{\mathrm{c}} \end{aligned} \tag{10}$$

where ρ_{w}, ρ_{c}, and ρ_{m} denote the densities of water, CO_2, and the local mixture, respectively (kg/m^3); μ_{w}, μ_{c}, and μ_{m} denote the corresponding dynamic viscosities (Pa(s).

Boundary-Condition and Numerical-Simulation Procedure

The core is initially assumed to be fully water-saturated and water-wet. CO_2 (green in Fig. 4a) is injected from the left boundary at a prescribed volumetric flux q_{in}. Driven by the joint action of pressure gradient, capillary force and viscous resistance, the CO_2 front advances through the core, displacing the water phase (purple in Fig. 4b) until breakthrough or residual water saturation is reached, after which production ceases at the outlet (q_{out}).

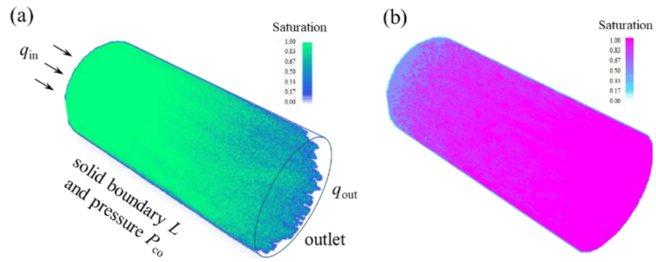

Figure 4—(a) schematic diagram of model boundary and CO_2 invasion, (b) water saturation distribution.

Boundary conditions were prescribed as follows. A constant-velocity inlet was imposed on the left boundary, q_{in}, while the right boundary was assigned a zero normal pressure gradient with a fixed boundary L. All pore-wall surfaces satisfied the no-slip condition and retained their specified water-wet character. The remaining external faces were held at a constant pressure P_{co}.

Two sets of fluid properties were considered. Under reservoir conditions (15 MPa, 354 K) the $scCO_2$ has a density of 395 kg/m^3 and a viscosity of 0.06 mPa(s (Yang et al., 2023), whereas water has a density of 975 kg/m^3 and a viscosity of 0.38 mPa(s; the interfacial tension is 0.03 N/m and the contact angle is 160°. Under laboratory conditions (0.1 MPa, 298 K) CO_2 behaves as a gas with a density of 1.8 kg/m^3 and a viscosity of 0.015 mPa(s (Span and Wagner, 1996), while water possesses a density of 998 kg/m^3 and a viscosity of 1.0 mPa(s; the corresponding interfacial tension and contact angle are 0.07 N/m and 160°, respectively.

Advancing the interface under the above boundary and property constraints yields a coupled, highly nonlinear algebraic system for the volume fraction, velocity, pressure and saturation fields. After spatial discretisation the system is solved iteratively until convergence. To cope with the very large grid size, GPU-based parallel computing (Naumov & Arsae, 2015) combined with an algebraic multigrid accelerator (Brandt, 1986) is employed, delivering a substantial reduction in wall-clock time.

Model Validation

The permeability K of the digital core was first calibrated by simulating a single-phase water-flood test, following exactly the same workflow used in laboratory core-plug measurements. Details of the numerical procedure can be found in Sun et al. (2017) and Chen et al. (2016). Under the experimental boundary conditions, the simulated injection-production pressure drop and outlet flow rate were post-processed with Darcy's equation to obtain the absolute permeability. As listed in Table 1, the computed values agree closely with the laboratory measurements.

Table 1—omparison of characteristic points on the simulated and measured relative-permeability curves

Core No.	Core type	K, mD exp./sim.	P_{inj}, MPa exp./sim.	Swi exp./sim.	K_{rg} exp./sim.	K_{m} exp./sim.	S_{gm}, % exp./sim.
L85	porous	1.35/1.66	1.52/1.52	46.77 / 45.45	0.37 / 0.37	0.11 / 0.93	70.03 / 68.24
L59	vuggy	12.32/14.11	0.52/0.52	32.31 / 32.99	0.60 / 0.58	0.12 / 0.10	76.15 / 73.73
L27	fractured	123.51/118.4	0.15/0.15	49.13 / 46.25	0.74 / 0.73	0.19 / 0.14	69.16 / 71.52

Subsequently, the CO_2-water relative-permeability experiment was reproduced numerically in strict accordance with the protocol of Shen et al. (2019). The relative permeabilities for core plugs L85, L59 and L27 were calculated with the Jones and Roszelle (1978) method, using the same injection pressure P_{inj} as in the corresponding physical tests (Table 1). Figure 5 compares the simulated and experiment measured relative-permeability curves. Key characteristic parameters—residual water saturation S_{wi}, CO_2 relative permeability at S_{wi} (k_{rg}), crossover relative permeability k_m and crossover water saturation S_{gm}—are summarised in Table 1. The close match between simulation and experiment confirms the fidelity of the reconstructed core models and the reliability of the numerical methodology employed in this study.

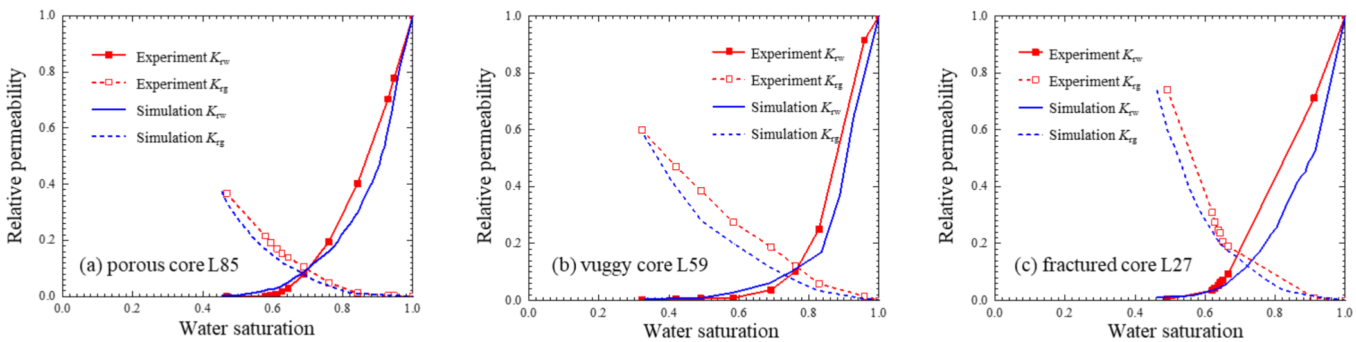

Figure 5—Comparison of simulated and experimental CO_2–water relative-permeability curves for various core types under laboratory conditions.

Although the overall agreement is good, the simulated results do not reproduce the experimental data exactly and several discrepancies remain, which can be attributed to the following factors: (1) Limitations in micro-CT resolution prevent the explicit extraction and representation of nano-scale pore throats; these throats are therefore homogenised in the digital core with a single, low-permeability constraint; (2) Conventional two-phase flow theory neglects geomechanical coupling and the presence of connate water films on mineral surfaces, both of which can influence phase conductivities; (3) Fluid-property inputs such as gas viscosity are estimated with empirical correlations whose idealised assumptions deviate from real behaviour.; (4) The laboratory measurements themselves are subject to experimental uncertainty.

Results and discussion

Comparison of two-phase relative-permeability curves for cores with different pore structures

Figure 6 compares the Two-phase relative-permeability curves of three different cores, representing porous core (L85), vuggy core (L59) and fracture core (L27), obtained under reservoir and laboratory conditions. The results demonstrate that changes in temperature and pressure systematically shift both the shape and the position of the curves.

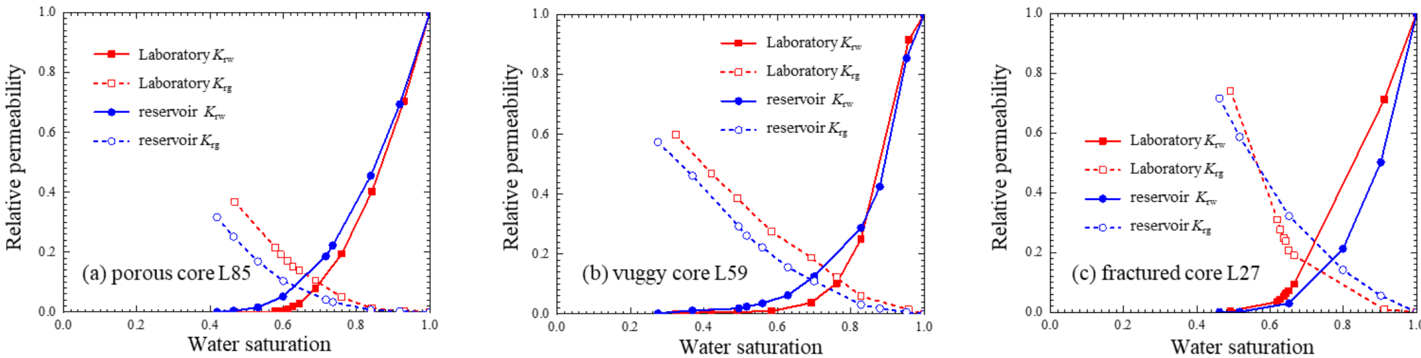

Figure 6—Two-phase relative-permeability curves for different core types under reservoir versus laboratory conditions.

Porous core L85. Under reservoir conditions the irreducible water saturation is lower and the corresponding carbon dioxide relative permeability is reduced; the crossover point moves to lower water saturations, enlarging the two-phase flow window by about 0.05. Both the water–phase decline and the carbon dioxide-phase rise in relative permeability are more gradual. Vuggy core L59. A similar trend is observed: lower irreducible water saturation, lower carbon dioxide relative permeability at this saturation, and a leftward shift of the crossover point, resulting in a 0.06 increase in the two-phase flow interval. Fracture core L27.Reservoir conditions again reduce irreducible water saturation and carbon dioxide relative permeability at that saturation, but the crossover point shifts slightly to higher water saturations. The two-phase flow window expands by only 0.02, less than in the other two cores.

Among the three cores, the vuggy core L59 exhibits the widest two-phase flow range and therefore the highest displacement efficiency, owing to the abundance of large solution voids that contain easily mobilised fluids. In contrast, the fractured core L27 has the narrowest two-phase window, the poorest displacement efficiency and the weakest concurrent flow capacity; once CO_2 breakthrough occurs, further improvement in sweep efficiency is limited.

CO_2 invasion patterns under varying conditions

The CO_2-water relative-permeability curves measured at different temperature–pressure regimes (Figure 6) show clear discrepancies: the two-phase flow window obtained under in-situ reservoir conditions is consistently broader than that derived under ambient laboratory conditions. To clarify this behaviour, the present section illustrates the CO_2 invasion patterns observed in three core types at both conditions and analyses the migration and distribution of CO_2/scCO_2 and water during displacement. Figures 7, 8 and 9 display the two-phase distributions at CO_2 breakthrough and at the residual-water stage for the matrix-dominated core L85, the vuggy core L59 and the fractured core L27, respectively; green represents CO_2 and purple denotes water. Porous core L85 and vuggy core L59. The simulated fluid distributions differ markedly between ambient and high-temperature/high-pressure (HTHP) reservoir conditions. Under HTHP, water viscosity decreases while CO_2 becomes super-critical and its viscosity increases, leading to a much larger viscosity ratio between the invading and displaced phases than at room conditions. A higher viscosity ratio suppresses viscous fingering of the invading fluid (Holtzman & Segre, 2015) and promotes a more compact displacement front (Lenormand et al., 1988). Consequently, sweep efficiency improves, the two-phase flow window widens and the crossover point of the relative-permeability curves shifts to lower water saturations. Because the vugs store larger volumes of mobile fluid that are readily produced, the vuggy core exhibits the widest two-phase window.

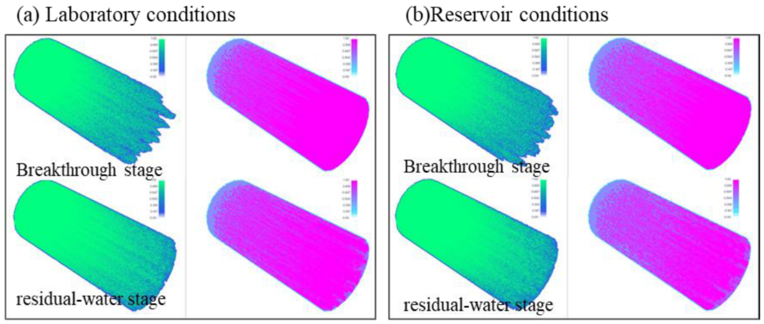

Figure 7—Comparison of two-phase fluid distributions at breakthrough and at residual-water stages (porous core L85)

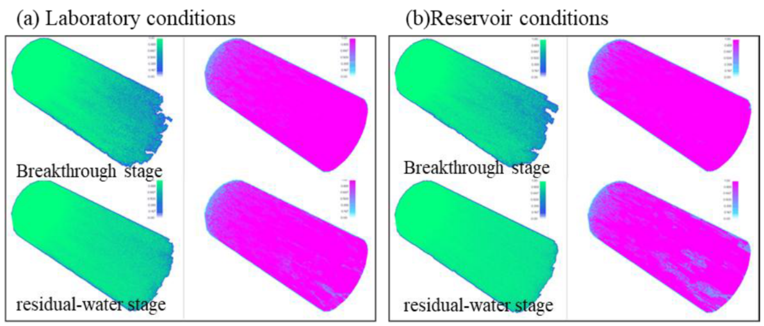

Figure 8—Comparison of two-phase fluid distributions at breakthrough and at residual-water stages (vuggy core L59)

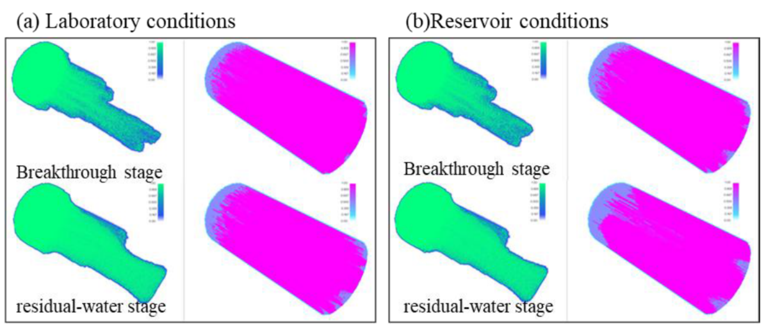

Figure 9—Comparison of two-phase fluid distributions at breakthrough and at residual-water conditions (fractured core L27)

Fracture core L27, shown in Figure 9, invading CO_2 first fills the fractures, rapidly expelling the fracture-bound water and causing a steep drop in water relative permeability. Owing to the pronounced heterogeneity, displacement is largely confined to the fracture network; water in the matrix pores is hardly obilized, leading to the highest residual-water saturation and the narrowest two-phase window among the three samples. The fluid distributions under ambient and HTHP conditions are very similar, indicating that in the presence of dominant fracture pathways, temperature, pressure and the two-phase viscosity ratio exert only minor influence on CO_2 displacement; accordingly, the relative-permeability curves change little.

Integrating the permeability data with the invasion images reveals that the post-displacement CO_2 permeability recovery varies with rock fabric: it is highest for the fractured core (largest CO_2 relative permeability at residual conditions), lowest for the matrix-dominated core, and intermediate for the vuggy core. The fractured sample flows mainly through fractures and dissolution seams—water in these channels is almost completely removed despite the overall low sweep efficiency. In the matrix-dominated sample, small throats generate high capillary forces that restrict CO_2 mobility and limit permeability recovery; the vuggy fabric falls between these two extremes.

Conclusions

High-resolution micro-CT models of porous-matrix, vuggy and fractured rocks were coupled with a GPU-accelerated VOF simulator to quantify CO_2/scCO_2-water displacement. Results confirm that pore-scale heterogeneity is the dominant control on flow. Fractured cores exhibit rapid fracture-dominated breakthrough with <30 % sweep, whereas vuggy cores sustain a uniform front and >60 % sweep; porous cores behave intermediately. Elevating pressure/temperature to 15 MPa/354 K increases scCO_2 viscosity and lowers that of water, enlarging the two-phase flow window and enhancing sweep in porous and vuggy media, but hardly affecting fracture-controlled systems. Thus, internal structure, rather than fluid properties, dictates displacement efficiency, while supercritical CO_2 delivers a stabler front and superior microscopic sweep—insights that should guide injection rate, volume and zonal-control strategies in CO_2 sequestration and EOR operations.

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